Building and evaluating a cross-hazard machine-learning approach to unified lunar forecasting

Luka Takki[a,*]

[a]European School of Luxembourg I, 7 Rue Simone de Beauvoir, L-1283 Luxembourg, Luxembourg

Abstract

LIPAS (Lunar Impact and Predictive Analysis System) is a hybrid forecasting and decision-support platform that combines machine learning with physics-based computational models to predict the Moon's space-weather conditions. These conditions include temperature, radiation, dust activity, micrometeorite impacts, solar storms, and moonquakes. By unifying space-weather data products from NASA and the National Oceanic and Atmospheric Administration (NOAA) alongside OMNI2 data, the LIPAS platform processes the data through a temporal Long Short-Term Memory (LSTM) neural network. Users can view the LIPAS space-weather predictions via an interactive, web-based dashboard for any lunar location.

This study addresses three research questions. First, it documents how the hybrid LIPAS model was developed and what was learned in that process. Second, it examines whether a unified multi-hazard system can support operational decision-making in a controlled mission simulation used alongside existing public NASA and NOAA products. Third, it tests whether the locked hybrid improves prediction for selected hazards relative to selected replicated public baselines and physics-only references. The quantitative evaluation now includes two 28-day 2026 windows and a disturbed 14-day historical window (May 2024), under an updated serve policy with storm-aware adaptive gating; the locked train-2 weights are unchanged. The contribution evaluated here is a hazard-specific hybrid architecture inside a lunar decision-support framework; LIPAS is assessed as a forecasting tool used alongside specialist NASA and NOAA models, not as a replacement for them.

Keywords: lunar environment, space weather, hybrid forecasting, machine learning, long short-term memory, mission decision support

1. Introduction

Renewed international interest in lunar exploration, particularly through NASA's Artemis program and its international partners, has increased the need for reliable and accessible information about the lunar environment (NASA, 2020). The Moon has no protective atmosphere or global magnetic field, leaving its surface exposed to large temperature variations, radiation, micrometeorite impacts, electrostatic dust, and seismic activity. These hazards can affect crew safety and the operation of spacecraft and surface equipment. A range of scientific models already exists for studying individual lunar hazards. These models, however, are generally designed for specific hazards and operate as separate tools. Their outputs are difficult to combine into a single operational view or use together for forecasting. Mission planners may therefore need to consult several data sources and models when assessing environmental conditions, which complicates decision making. The Lunar Impact and Predictive Analysis System (LIPAS) was developed to address this fragmentation by bringing multiple lunar-hazard forecasts into a single decision-support platform. The system combines machine-learning predictions with physics-based models, public heliophysics data[1], and adaptive weighting between the two approaches. The literature reviewed here describes specialized lunar hazard models and archives, while LIPAS explores how these sources can be combined with machine learning in a unified decision-support framework.

This study addresses three research questions:

*Corresponding author
Email address: lukatakki123@gmail.com (Luka Takki)

[1]Heliophysics studies the Sun, the solar wind, and their interaction with planetary environments throughout the heliosphere.

RQ1: Can a hybrid lunar-forecasting system be developed by effectively combining machine-learning and physics-based approaches?

RQ2: Can integrating multiple lunar environmental hazards and data sources into a unified forecasting and decision-support system support operational decision-making alongside existing public products?

RQ3: Does the locked LIPAS hybrid improve prediction performance for selected lunar hazards relative to selected replicated public baselines and physics-only references?

2. Overview of current models

2.1. General overview

The study of lunar environmental conditions has historically relied on single-purpose models, each focused on a particular hazard. Although these models incorporate advanced scientific understanding in their respective domains, they remain limited when applied to operational mission planning because they do not provide integrated insights or unified predictive support. Each model was built to solve a specific scientific problem rather than to operate as part of a broader forecasting system. This creates a gap because the interactions among hazards cannot readily be analyzed.

2.2. NASA DONKI model

The NASA Database of Notifications, Knowledge, Information (DONKI) API provides near-real-time monitoring of solar flares, coronal mass ejections (CMEs), and solar energetic particle (SEP) events (NASA, 2023b). DONKI is valuable as a space-weather event catalog for radiation context, but it is event-driven and primarily reactive, with limited temporal coverage and incomplete data for continuous mission planning. It does not itself produce multi-day forecasts or integrate other hazard types, making it insufficient as a stand-alone planning tool.

2.3. Temperature models

Temperature predictions rely heavily on the Diviner Lunar Radiometer Experiment (Paige et al., 2010; Williams et al., 2017; NASA, 2023a), which provides detailed infrared measurements of surface temperature across the Moon. These measurements accurately characterize lunar thermal conditions, including uneven terrain, but do not themselves produce short-term forecasts. Apollo-era heat-flow measurements are similarly sparse and lack the temporal resolution required for short-term forecasting (Langseth et al., 1972). Neither source was designed to predict future temperature; both principally describe measured or reconstructed conditions. The early prototype used a simplified Diviner-style dayside surface-temperature relation, T = 392 cos0.25(z) -273, (1) where T is surface temperature in degrees Celsius and z is the solar zenith angle at the target coordinates. Legacy nightside profiles use polynomial fits derived from Apollo heat-flow measurements. Although deterministic, these stand-alone relations do not dynamically adjust to changing space-weather conditions.

2.4. Lunar dust models (LADEE/LDEX)

The lunar dust environment has been studied by the Lunar Atmosphere and Dust Environment Explorer's Lunar Dust Experiment (LADEE/LDEX), which documented dust-particle density and electrostatic activity (Horanyi et al., 2015; NASA Planetary Data System, 2015). LDEX is important for understanding surface hazards, but its data are limited to the LADEE mission epoch and orbital coverage, providing neither continuous global monitoring nor direct predictive capability. The mission ended in 2014, leaving substantial temporal and spatial gaps.

2.5. Moonquakes and Apollo PSE

Moonquake monitoring relies on the Apollo Passive Seismic Experiment (PSE) catalog. Recent reanalysis identified thousands of previously undetected moonquakes (Onodera, 2024). The PSE dataset is retrospective, cannot provide real-time forecasting, and does not integrate other hazards. Nevertheless, it contains statistical patterns tied to the lunar tidal cycle. LIPAS uses these patterns as physics/catalog context for moonquake-rate estimates. Because deep moonquakes occur in clusters correlated with tidal stress between Earth and the Moon, recurring catalog patterns can be projected forward statistically even without real-time seismic sensors (Kawamura et al., 2017; Lammlein, 1977; Nakamura, 2005; Nunn et al., 2020).

2.6. Micrometeorite prediction with MEM 3

NASA's Meteoroid Engineering Model (MEM 3) estimates meteoroid collision rates (Moorhead et al., 2020). It is precise within its engineering domain but does not directly incorporate secondary operational effects such as dust lofting or the interactions of meteor-shower timing with surface operations. LIPAS therefore applies shower-calendar modulation and dust coupling to the MEM 3 baseline used in its physics path (American Meteor Society, 2026; International Meteor Organization, 2026; Szalay and Horanyi, 2015).

2.7. GOES-18 radiation context

NOAA's Geostationary Operational Environmental Satellite 18 (GOES-18) continuously monitors solar X-ray flux and

energetic-particle counts (National Oceanic and Atmospheric Administration, 2023). GOES measurements are made in Earth orbit and must be adapted for lunar-surface conditions because the Moon lacks a global magnetosphere. GOES-18 does not provide a dedicated multi-day lunar radiation forecast and was not designed specifically for lunar missions.

2.8. Summary

Individual models and archives such as DONKI, Diviner, MEM 3, PSE, and LDEX are essential for scientific research, but their limitations in coverage, integration, and predictive usability motivate a unified system. LIPAS combines the strengths of these single-purpose approaches with real-time NASA and NOAA space-weather data, historical lunar measurements, and a hybrid machine-learning architecture. The resulting framework is proactive and integrative, providing a comprehensive interface for mission planning and safety assessment while retaining the limitations of its source data and models.

3. System overview: LIPAS

3.1. Prototype limitations and migration to a backend

The original LIPAS prototype was a browser-based application in which data processing and machine learning were performed on the client side. As it grew, forecasts for physically constrained hazards such as temperature varied between training runs. This showed the need to retain stronger physical relationships and move processing to a backend that could support a more stable data and forecasting service. The application was therefore re-engineered as a server-assisted hybrid model with a Python backend. Data filtering, API polling, model training, and inference are handled on the server, which supplies a stable data service to the front-end dashboard. The current system can assign different levels of influence to machine learning and physics according to the hazard and current conditions.

3.2. Hybrid forecasting and uncertainty architecture

The core forecasting model uses an 18-feature input representation that contains hazard-related, temporal, spatial, and space-weather variables. Rather than processing only current conditions, the model receives a 72-day temporal window (one step equals one calendar day). It produces six target outputs: solar activity, radiation, temperature, moonquake rate, micrometeorite flux, and dust density. The larger input representation allows information from one environmental channel to contribute to the prediction of another. The temporal trunk is based on a long short-term memory architecture (Hochreiter and Schmidhuber, 1997). A physics-estimation function first calculates a physically grounded value for each hazard. A subsequent refinement step combines that value with the neural-network output, allowing machine learning to provide a correction instead of learning the complete physical relationship. For hazard channel i, the resulting prediction is $\hat{Y}_{\text{hybrid},i} = w_i \hat{Y}_{\text{ML},i} + (1 - w_i)\hat{Y}_{\text{physics},i}$. (2) The weight is evaluated from the current space-weather regime and the neural network's uncertainty: $w_i = \max\{0, \min[1, (w_{\text{base},i} + \Delta_{\text{regime},i}) \times \max(0.35, 1 - 1.6\sigma_{\text{ML},i}/S_i)]\}$. (3) Here wbase,i is the baseline machine-learning weight, $\Delta_{\text{regime},i}$ is an adjustment for the current storm posture, $\sigma_{\text{ML},i}$ is the neural uncertainty estimate, and S i is the physical operating span of the channel. Uncertainty is estimated with Monte Carlo dropout: the network is evaluated repeatedly while a randomly selected subset of neural connections is disabled on each pass. Closely grouped outputs indicate a stable learned prediction, whereas a wider spread indicates greater uncertainty in the learned model.[2] When $\sigma_{\text{ML},i}$ rises, Eq. (3) reduces the neural contribution and shifts the forecast toward the physics baseline. Where physics provides a strong basis, LIPAS assigns greater weight to the physics path. In the locked production blend, temperature, moonquakes, micrometeorite flux, and dust are predominantly physics-led, while solar activity retains a machine-learning weight of 0.4. The served baseline sets radiation to physics-only (0.0). Storm-aware adaptive gating additionally zeros the solar and radiation machine-learning weights when the space-weather regime is storm-like or when the live radiation input is at or above 0.20 mSv, routing those channels through the physics path under disturbed conditions. The backend also supports optional ensemble forecasts for uncertainty estimation, although the evaluated production path uses a single locked temporal model.

3.3. Development of the hybrid framework

The current LIPAS implementation uses HTML, CSS, and JavaScript for the front end and a Python/TensorFlow backend for data ingestion, model training, physics blending, and forecast serving (Google

[2] This is the uncertainty that comes from the model, or from gaps in the training data.

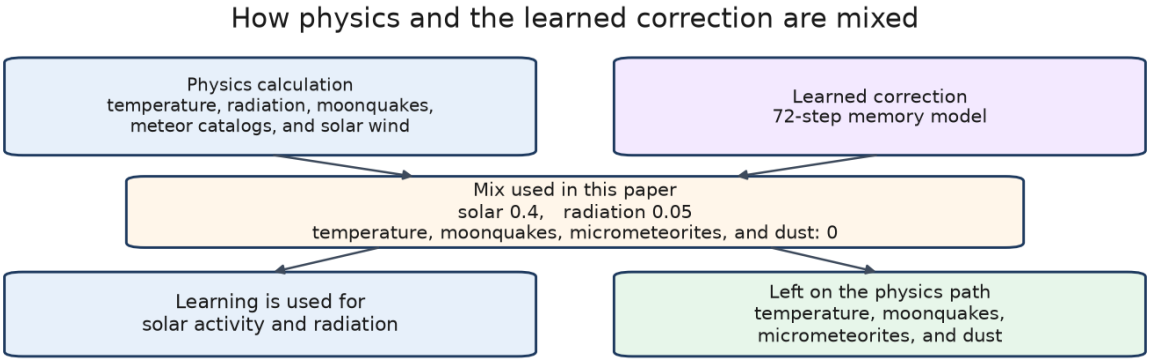


Figure 1: Locked hybrid mix. The learned correction is used for solar activity (weight 0.4). Radiation uses physics in the locked serve policy (baseline weight 0.0, formerly 0.05). The other four hazards stay on the physics calculation.

TensorFlow Team, 2026; Chollet et al., 2015). Multiple production training runs were completed. The reported quantitative evaluation used the second training-run model because it was the best-performing completed production model when the model was locked for testing. Its chronological held-out test RMSE was approximately 4.97, compared with approximately 13.66 for the corresponding persistence baseline; no later completed run had surpassed it at the evaluation cutoff. Earlier prototype experiments used the browser-based neural network that preceded the Python hybrid backend. They are treated as developmental diagnostics rather than evidence of current model performance because they lacked the architecture and baseline framework used in the reported evaluation. The locked production model was subsequently evaluated using the framework in Section 4.

3.4. Data ingestion and pipeline orchestration

The backend handles real-time data streams through a modular ingestion pipeline that queries, cleans, and normalizes external APIs into the 18-feature representation. The pipeline joins CME events from DONKI, solar indices from the NOAA Space Weather Prediction Center, and interplanetary magnetic-field data from OMNI (NASA, 2023b; National Oceanic and Atmospheric Administration, 2023; NASA Goddard Space Flight Center, 2026; King and Papitashvili, 2005). These space-weather feeds are combined with LRO Diviner thermal maps, CRaTER radiation logs, and LADEE/LDEX dust measurements (NASA, 2023a,c; NASA Planetary Data System, 2015). Long-term baselines include Apollo PSE data, Apollo heat-flow measurements, MEM 3, and peer-reviewed micrometeorite-flux data. LIPAS is a hybrid forecasting system in which machine learning is used only where it can improve on an established physics calculation. Where a physics relation already describes a hazard well, that relation remains the primary estimate, and the learned correction is down-weighted or switched off (Section 3.2).

3.5. Command-center dashboard and map visualization

The user interface was overhauled after the prototype limitations were identified. The resulting mission-control-style interface uses CSS Grid and Flexbox and consists of a dashboard, mission planner, and Cesium globe (Cesium GS, 2026; Chart.js Contributors, 2026). The dashboard includes more than 50 named lunar locations and a coordinate search interface for arbitrary sites. It provides topographic maps with hazard overlays, 30day outlooks, particle-flux visualizations, and warnings and recommendations for mission planning. Historical and predicted trends can be displayed as line, bar, or area charts. The missionplanning interface accepts mission parameters and duration targets. Based on available forecasts, it proposes EVA departure windows, assesses landingzone conditions, and generates an exportable mission plan. A Luna Assistant provides additional hazard information. These dashboard and planning views are the operator-facing surface of LIPAS. The Cesium page supplies an interactive two or threedimensional lunar map with time-dependent overlays for dust, radiation, and solar drivers. Together these views form the redesigned LIPAS operator interface used after the prototype migration. Figure 3 shows the operator interface in four panels: site selection, diagnostics, exportable mission reporting, and the Cesium globe. All quantitative window, mission, and ablation scores in this paper use the locked production model; the April prototype diagnostic remains separate.

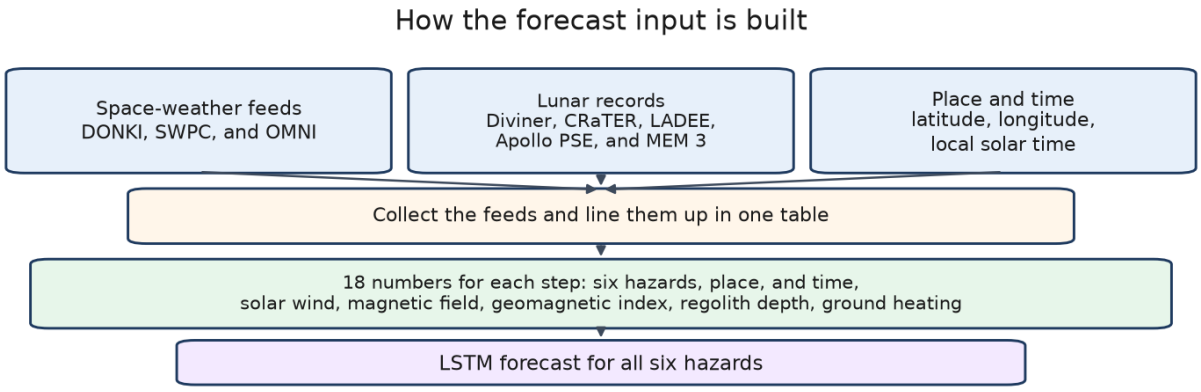
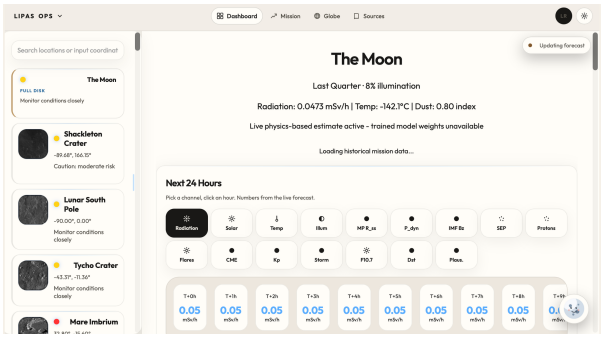
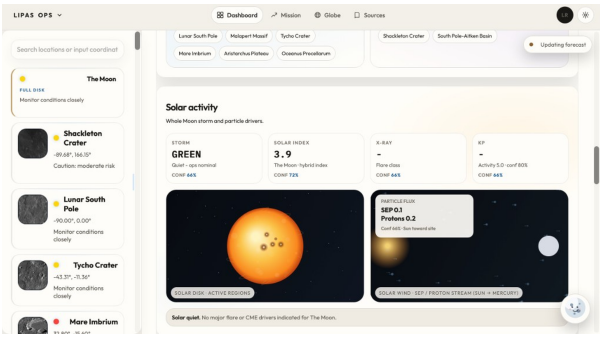


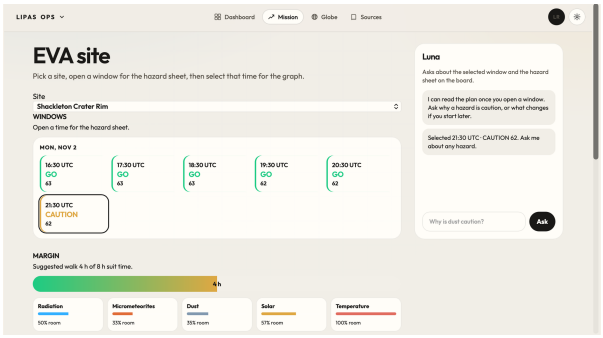
Figure 2: Data path into the forecast. Public feeds, lunar records, and the place and time become the 18 numbers the model reads.



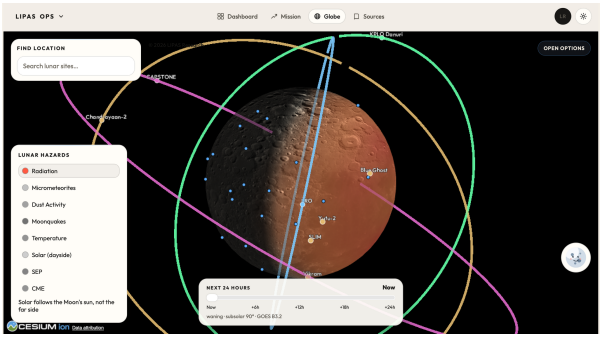
(a) Site selection, coordinate search, and topographic hazard map.



(b) Forecast drivers, status, and diagnostics.



(c) Exportable mission report and risk summary.



(d) Interactive lunar globe and hazard overlays.

Figure 3: Representative LIPAS operator workflow. Dashboard: https://lipas.netlify.app.

4. Methods

4.1. Experimental design and evaluation framework

The research was divided into three phases, each addressing one research question. The first focused on developing LIPAS and identifying lessons from that process, including the April prototype diagnostic. The second examined operational use in a simulated mission. The third evaluated whether the locked hybrid system improved predictive performance relative to established physics-based and agencystyle reference approaches. All reported quantitative evaluations in Windows A, B, and C, the mission simulation, and the ablation used the locked production model from the second training run. The April prototype scores are separate and are not pooled with those locked-model results.

4.1.1. Phase 1: development of the hybrid model

The first phase examined the transition from the browser prototype to the hybrid system. It included a daytime prototype evaluation at 0°N, 0°E from 5 to 19 April 2026. The purpose was not to establish operational accuracy, but to identify weaknesses in the client-side machine-learning approach and determine which channels required a physics-based foundation. Treating all six hazards as machine-learning problems produced unstable results for strongly physics-constrained hazards. Consequently, temperature, moonquakes, dust, and micrometeorites were kept predominantly physics-based, while machine learning was given a larger role in solar activity and radiation. This phase also established the 18-feature input structure (hazard, time, place, and space-weather numbers), the 72-day temporal window, the physics_estimate paths (Sections 3.2 and 4.4), and adaptive hybrid weighting (Eqs. (2)–(3)). These components were combined into the locked model used in the remaining phases. Continuous co-located measurements are unavailable for all six hazards, so comparison targets are called evaluation labels rather than ground truth unless a direct observation exists (Table 1). Performance used RMSE and MAE. Site geometry for Windows A and B is shown in Figure 4.

4.1.2. Prediction setup and leakage controls

The locked training rows are daily: one time step is one calendar day. Each training example reads the preceding 72 days of the 18 input features and predicts constructed hazard labels for the next calendar day relative to the final day in that window. At prediction time the model may use only information dated on or before that final day. The target columns are not raw co-located surface measurements; they are next-day labels built from the same public drivers and physics paths used elsewhere in LIPAS (Table 1). Because several drivers change slowly, and because some input channels are related to the quantities used to build the labels, part of the learned skill can look like short-horizon forecasting and part can look like reconstruction of a smooth proxy series. The paper therefore treats the locked hybrid as a one-day-ahead prediction of constructed labels, not as proof of multi-day storm foresight. Short gaps in the driver series were filled by limited linear interpolation before training; feature scaling used means and standard deviations fitted on the training block only and then applied to validation and test. No separate resampling of the locked daily grid was used.

Chronological separation was used to limit data leakage. For each site, rows were ordered by date and cut into training, validation, and test blocks (about 70%, 15%, and 15%), so later dates never entered an earlier block. The eighteen input features were fixed in the model design and were not chosen using the validation or test blocks. Overlapping 72-day windows are formed with a stride of seven days inside a split, but they do not cross the train, validation, or test date cuts, and later observations are not written into earlier inputs. The same multi-output split is used for all six hazards. Row counts and date ranges are listed in Appendix Appendix E. Windows A, B, and C are separate locked evaluation periods for the served hybrid against the study labels on those dates and are reported apart from the chronological hold-out test.

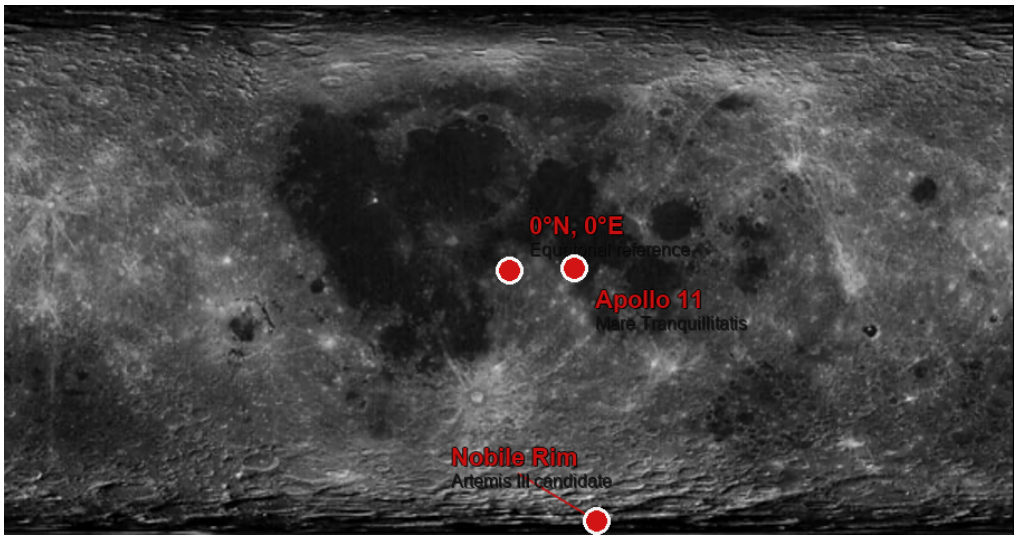


Figure 4: Lunar locations used in the reported studies: the equatorial reference (0°N, 0°E), Apollo 11, and Nobile Rim.

Predictions in the locked windows were compared with persistence, physics-only references, and replicated public baselines (Section 4.4), replayed from archived OMNI, NOAA, GOES, and DONKI data.

4.2. Qualitative comparison criteria

Before the quantitative windows, LIPAS and selected agency models and archives are scored on nine ordered criteria. Labels run from None to Very High and are not converted into a numeric total or summed rank. Appendix B records the scoring rules; Table 2 summarizes the matrix. The locked-model RMSE and MAE studies remain the primary quantitative evidence for research question 3. *Hazard coverage.* None: no hazard addressed; Very Low: one hazard; Low: up to two hazards; Moderate: three to four hazards; High: five hazards; Very High: six or more integrated hazards. *Data-fusion capability.* None: one dataset; Low: multiple datasets without integration; Moderate: partial integration; High: integrated multi-source system; Very High: real-time dynamic multi-source fusion. *Temporal forecasting ability.* None: archival only; Low: event-driven alerts without a time horizon; Moderate: short-range estimates for a narrow domain; High: multi-day forecasting for one or more hazards; Very High: multi-day forecasting across all hazards. *Spatial coverage.* None: no spatial resolution; Very Low: specific historical sites; Low: one region or polar area; Moderate: broad regional coverage; High: global or near-global coverage; Very High: complete lunar-surface coverage with location selection.

Realtime operational capability. None: historical only; Low: archival access with significant delay; Moderate: near-real-time with limited updates; High: live feeds with regular updates; Very High: continuous real-time data with automated processing. *Robustness and stability.* Low: limited coverage or known gaps; Moderate: functional with notable external dependencies; High: well-validated and stable in a narrow domain; Very High: continuously calibrated mission-grade instrumentation with a long operational history. *Validation strength.* None: no comparison; Low: informal or limited comparison; Moderate: structured comparison over a limited period; High: peer-reviewed validation against direct measurements; Very High: direct physical measurement, self-validating by design. *Usability and operational readiness.* Very Low: raw scientific data requiring specialist processing; Low: technical interfaces; Moderate: accessible API or web portal; High: clear operational interface; Very High: interactive dashboard for non-specialist mission planners. *Openness and accessibility.* Low: restricted or proprietary; Moderate: archives requiring registration or processing; High: open API or published datasets; Very High: fully open browser-based access without setup. No aggregate result in this paper represents validation against simultaneous in-situ measurements of all six hazards across the lunar surface. An ablation analysis later replaces selected inputs with Window A means while holding the locked model fixed (no retraining). Baseline configurations and hazard-specific replication continue in Sections 4.3–4.4.

Performance was evaluated primarily by root mean square error (RMSE) and mean absolute error (MAE). The Window B micrometeorite comparison used the MEM 3 seasonal background with American Meteor

Society (AMS) shower modulation for LIPAS, while the replicated comparison omitted that additional shower term. This tests the effect of explicitly representing shower timing during the Perseid period, not general superiority over NASA's complete meteoroid-modeling framework. An ablation analysis held the locked model fixed and replaced one input, or a selected pair of inputs, with its Window A mean. Radiation and solar-activity RMSE were then compared with the full model (Tables 6 and 7). This is an input-sensitivity test, not an ML-only or physics-only pathway comparison.

4.2.1. Phase 2: operational mission evaluation

The second phase examined whether integrating several environmental hazards could support operational usability during a simulated lunar mission. A controlled radiation-focused EVA scenario used a timeline of 7 h 37 min, matching the duration of the second Apollo 17 EVA. LIPAS was monitored at fixed intervals and compared with information an operator would receive from separate public NASA and NOAA products, including GOES, NOAA SWPC, and DONKI. The same predefined caution and no-go thresholds were applied to each pathway. The comparison focused on when each system produced warnings and when the simulated EVA reached caution or no-go conditions. It did not establish that LIPAS is safer than NASA systems or measure usability with mission personnel.

4.2.2. Phase 3: quantitative prediction evaluation

The third phase evaluated the locked hybrid system against physics-based and replicated NASA/NOAA reference approaches at three locations: 0°N, 0°E, Apollo 11, and Nobile Rim (Fig. 4). Window A covered 17 June–14 July 2026 (predominantly mixed illumination, with an embedded radiation spike); Window B covered 18 July–14 August 2026 (predominantly lunar day); and Window C covered 4–17 May 2024 (a disturbed historical interval). For Window B micrometeorites, LIPAS used MEM 3 with American Meteor Society (AMS) shower-calendar modulation, while the replicated comparison did not apply that AMS shower calendar. Results then report the studies in matching order: prototype diagnostics, qualitative comparison, locked evaluation windows, mission simulation, and input ablation.

4.3. Baseline configurations and control environments

Reference predictions were obtained from public databases, peer-reviewed formulas, engineering-model exports, and agency documentation. In this paper, a replicated NASA/NOAA reference means an author-implemented reconstruction from those public materials; it does not mean that internal, proprietary, operational, or flight-certified agency software was executed. MEM 3 is treated separately because the comparison used an engineering-model export available to the project. Where an established product did not provide a direct multi-day lunar prediction for a study interval, the relevant public method was reproduced as closely as the documented equations and open data allowed. For every hazard, the study recorded the source, procedure, and whether the target was a direct measurement, engineering baseline, or physics proxy. In the mission simulation and ablation tests, coordinates and physical constants were held fixed while selected inputs were altered. The mission scenario used archived quiet conditions as a background and injected controlled changes, with status checked at fixed intervals. The ablation tests replaced selected driver or hazard inputs with their Window A means as sensitivity tests. Table 1 summarizes provenance for each evaluation target. Section 4.4 then specifies how each hazard baseline was replicated for Windows A and B, including the physics_estimate paths used when a public product did not already publish a multi-day lunar forecast for the study coordinates and dates. Where a channel mixes an engineering export with an author-added term (for example MEM 3 plus the AMS shower calendar), both pieces are stated explicitly so that a reader can replay the comparison without assuming access to internal agency software. Persistence and physics-only references share the same replay calendar and site list as the locked hybrid forecasts. Control environments therefore differ by which drivers are visible to each pathway, not by which dates or coordinates are evaluated. That separation is what lets the later Results tables attribute RMSE and MAE changes to the locked blend, the AMS term, or an ablated input rather than to a silent change of study geometry.

Channel	Principal target construction	What is directly observed	Validation status
Solar activity	Archived OMNI/SWPC indices and DONKI event context mapped to the study's solar-activity label	Solar-wind, field, flare, and geomagnetic drivers	Constructed label from measured drivers
Radiation	CRaTER-calibrated background combined with archived solar and proton-event terms	Orbital particle and space-weather measurements	Observation-constrained lunar-surface proxy
Temperature	Diviner/Apollo-constrained thermal equations evaluated at each site and time	Orbital temperature maps and historical heat-flow measurements	Observation-constrained physics proxy
Moonquakes	Apollo PSE catalog rates with tidal, terminator, and location adjustments	Historical Apollo seismic events	Historical-catalog proxy
Micrometeorites	MEM 3 engineering output with a separately stated AMS shower-timing term	MEM 3 output and terrestrial shower timing	Engineering-model/shower proxy
Dust	LDEX-informed background with impact, local-time, and terminator terms	Historical orbital dust detections	Observation-informed physics proxy

Table 1: Provenance and interpretation of the six quantitative evaluation targets.

4.4. Baseline replication and model specifications

4.4.1. Temperature

Temperature baselines used lunar thermal physics constrained by Diviner observations and Apollo nightside measurements (Vasavada et al., 1999; Paige et al., 2010; Hayne et al., 2017; Williams et al., 2017; Langseth et al., 1972). On the dayside, surface temperature was computed from the Stefan-Boltzmann balance $T = \left(\frac{(1-\alpha)I_0 \cos\theta}{\epsilon\sigma}\right)^{1/4}$, (6) where I0 is solar irradiance adjusted for Earth-Sun distance, α is surface albedo[3], ϵ is emissivity, σ is the Stefan-Boltzmann constant, and θ is the local solar-incidence angle. Nightside temperatures were calculated with an Apollo-based cooling curve and adjusted for latitude and a small interior heat term. Site reflectivity, emissivity, terrain, and thermal inertia were taken from the LIPAS site data. Where available, Diviner observations could further constrain the estimate. The early prototype also used Eq. (1); the reported studies used Eq. (6) and the Apollo-based nightside model. An equivalent form used in the implementation is (1 - α)Is = $\epsilon\sigma$T 4. (7)

4.4.2. Solar activity and radiation

Solar and radiation baselines were built from public space-weather data rather than a direct lunar radiation forecast. Solar activity used OMNI2 and SWPC quantities including Kp, the interplanetary magnetic-field component Bz[4], solar-wind speed and density, and F10.7 where available. DONKI flare, CME, geomagnetic-storm, and SEP records were included for the study dates. The radiation path used a CRaTER-based galactic-cosmicray background of approximately 0.38 mSv day-1, converted to an hourly value and adjusted for solar activity (Spence et al., 2010; Schwadron et al., 2014; NASA, 2023c). Additional quiet, storm, and SEP terms used Kp, southward Bz, high solar-wind speed, GOES proton measurements, and solar activity. A small latitude adjustment was applied near the lunar poles. GOES X-ray and proton measurements were model inputs, not direct lunar-surface radiation measurements.

4.4.3. Micrometeorites

Micrometeorite baselines were replicated from the project's MEM 3 export (Moorhead et al., 2020). The annual flux set the principal background level, and available highand low-density ranges represented seasonal variation. LIPAS added AMS-based meteor-shower modulation separately from the basic MEM 3 baseline. A latitude term was applied only in the LIPAS physics path. For Window B, the replicated comparison used MEM

3 without the AMS shower term, while LIPAS used MEM 3

plus shower modulation. This directly isolates the effect of meteor-shower timing.

4.4.4. Dust

Dust baselines used LADEE/LDEX data rather than a live global forecast. The model combined a normal background with dust associated with micrometeorite impacts, enhanced dust near terminators[5], an optional LDEX local time factor, and a small regolith depth effect[6] (Colwell et al., 2007; Horanyi et al., 2015; Stubbs et al., 2006). It is therefore a physics proxy based

[3] Albedo is the fraction of incoming radiation reflected by a surface; emissivity describes how efficiently that surface emits thermal radiation.

[4] The interplanetary magnetic field is carried outward by the solar wind; a sustained southward Bz component can indicate stronger coupling with Earth's magnetic environment.

[5] The lunar terminator is the moving boundary between illuminated day and darkness, where rapid electrical and thermal changes may affect dust behavior.

[6] Regolith is the loose layer of fragmented rock and dust covering the lunar surface.

on LDEX observations, not a claim of continuous dust measurement at the Artemis sites.

4.4.5. Moonquakes

Moonquake baselines used the Apollo PSE catalog. The model combined deep, shallow, and thermal moonquake rates, enhanced activity near terminators, a small equatorial preference for deep events, and a weak regolith effect. No real-time lunar seismometer network was assumed. The output is a catalog-based physics proxy rather than a direct measurement of current activity at the study sites. Persistence baselines assumed that the latest known value remained constant. Physicsonly references were generated from the common replay data. RMSE and MAE were calculated for LIPAS against persistence, physics-only paths, and each replicated reference baseline. The mission simulation used common LIPAS outputs and baselines throughout the Apollo 17-length timeline, and the ablation study used the same replay data while replacing selected inputs with their Window A means.

5. Results

5.1. Prototype diagnostics

The preliminary studies evaluated the browser prototype under daytime conditions at 0 deg N, 0 deg E from 5 to 19 April 2026. One study tracked all six hazards against weakly replicated NASA and NOAA baselines. Another compared temperature with a Diviner-style dayside relation. The aim was to see where the prototype failed under that weak-replica setup before the locked hybrid model was trained.

Figure 5 summarizes that April diagnostic across six hazards. It is not the locked window evaluation in Section 5.3. Temperature is the only hazard where the replica scores higher; the prototype is higher on radiation, micrometeorites, moonquakes, dust, and solar activity. The temperature miss is the actionable finding that motivated the physics-led temperature path in the locked model. Figure 6 shows the same miss as a daytime temperature trace against a Diviner-style dayside estimate. Section 5.2 then scores LIPAS against selected agency models and archives on the nine qualitative criteria defined in Section 4.2.

5.1.1. April temperature trace

The same April days, at 0°N, 0°E, are plotted below against a Diviner-style dayside estimate. Figure 6 shows the same miss: the prototype overshoots the Diviner-style estimate near peak heating. Figures 5 and 6 are the April check only; they are not repeated in Windows A, B, and C.

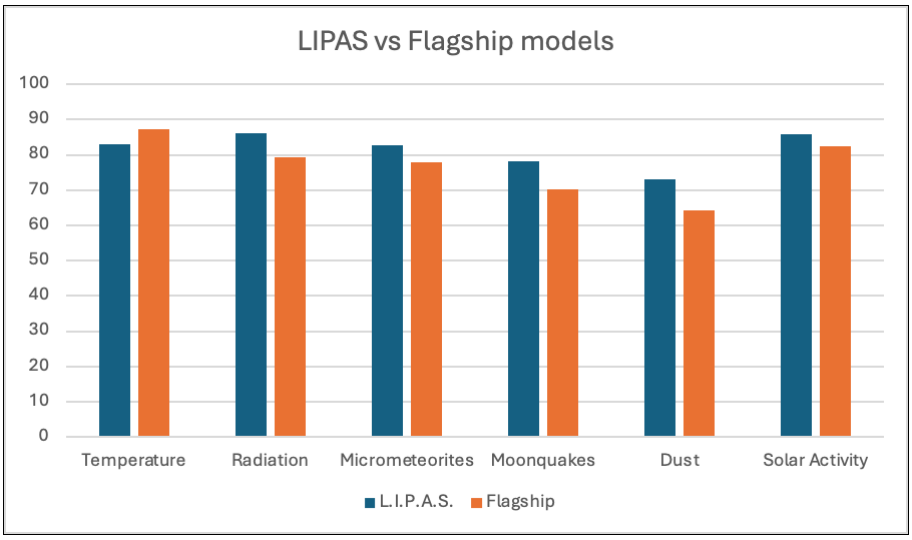


Figure 5: Preliminary LIPAS versus replica scores across six hazards (5 to 19 April 2026, daytime, 0°N 0°E).

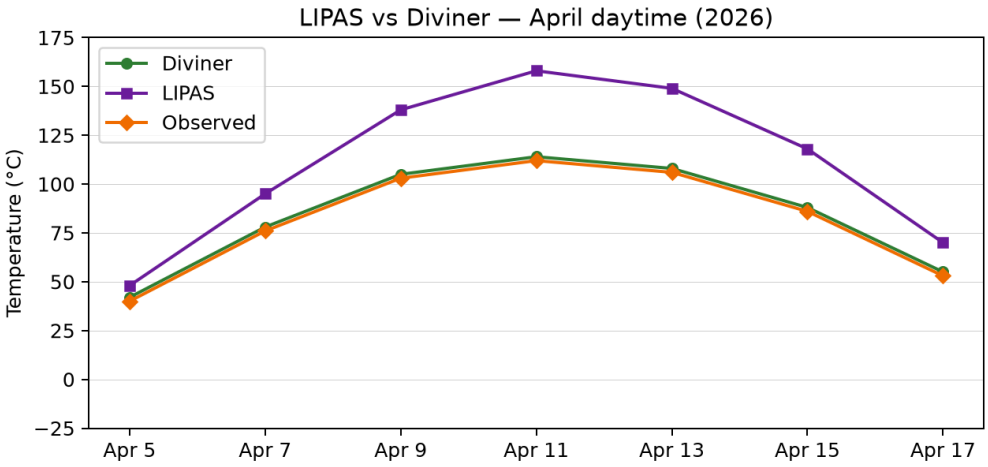


Figure 6: Daytime temperature at 0°N, 0°E (5–19 April 2026). The prototype overshoots the Diviner-style dayside estimate near peak heating.

5.2. Qualitative comparison

LIPAS and six reference systems were scored on the nine criteria in Section 4.2. DONKI monitors events, Diviner and LDEX archive measurements, Apollo PSE is a historical catalog, MEM 3 is a statistical engineering model, and GOES-18 is a real-time monitor used here from Earth orbit. Scores run from None to Very High and are not added up. VH = Very High; H = High; M = Moderate; L = Low; VL = Very Low; N = None. Specialist systems score high where they were built to be strong, and low on fusion and forecasting. LIPAS is scored high where the interface has to carry several hazards at once. The matrix is not a ranking, and it does not say that LIPAS is more accurate than a specialist archive.

System	Hazard	Fusion	Forecast	Spatial	Real-time	Robust.	Valid.	Usability	Open
LIPAS	VH	VH	H	H	H	M	M	VH	H
DONKI	L	L	L	M	H	H	H	M	H
Diviner	VL	L	L	VH	L	VH	VH	L	M
LDEX	VL	L	N	L	N	M	H	VL	M
Apollo PSE	VL	L	N	VL	N	H	VH	L	H
MEM 3	VL	L	M	H	L	H	H	L	M
GOES-18	L	M	M	M	VH	VH	VH	M	H

Table 2: Qualitative comparison of LIPAS with selected agency models and archives. Criteria follow Section 4.2. Labels are not summed. VH = Very High; H = High; M = Moderate; L = Low; VL = Very Low; N = None.

5.3. Locked evaluation windows

5.3.1. Window A, predominantly mixed

Window A covers 17 June to 14 July 2026 at 0°N 0°E, Apollo 11, and Nobile Rim (84 site-days). The interval is predominantly mixed in local solar time and includes an embedded radiation spike. The same blend, adaptive gating rules, and references are used at every site.

Solar activity shows a tiny hybrid gain (RMSE 14.15 versus 14.17). Radiation RMSE is slightly higher for the hybrid than the replica (1.264 versus 1.259). Temperature, moonquakes, and dust match because they share a physics path (39.9, 1.32, and 0.543). The micrometeorite channel uses blend weight zero and is not an independent win (0.0 versus 0.606 for the replica). Skill 96 = zero site RMSE, not a percent score.

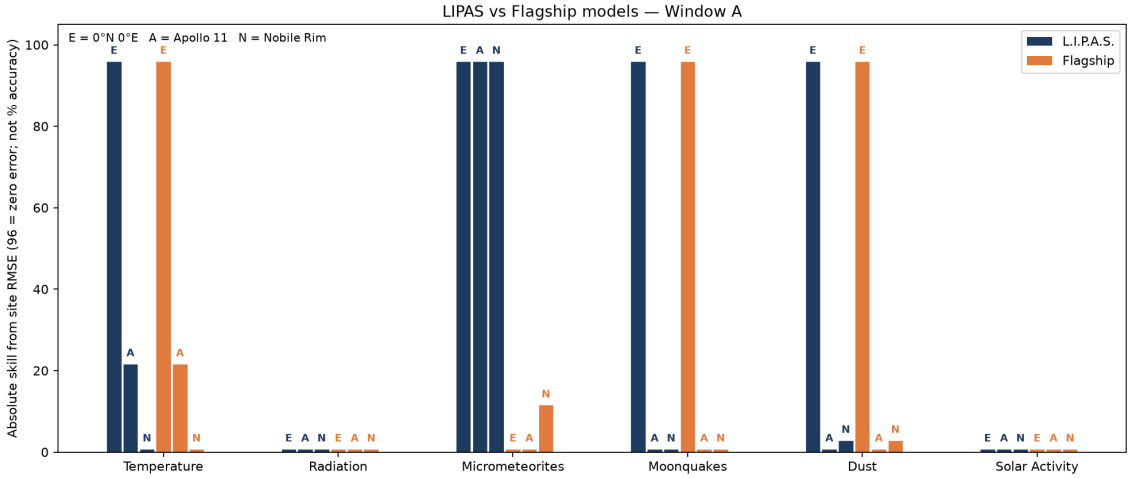


Figure 7: Window A skill from per-site RMSE (17 June to 14 July 2026; 84 site-days). E = 0°N 0°E; A = Apollo 11; N = Nobile Rim. Navy is LIPAS; orange is the replica.

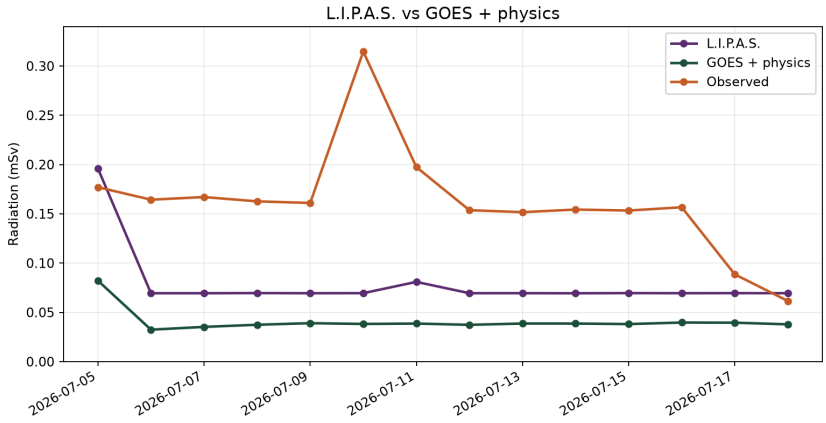


Figure 8: Window A radiation. LIPAS, the GOES-plus-physics reference, and the evaluation labels.

Hazard	Hybrid	Replica	Outcome
Solar activity	14.15	14.17	Hybrid lower by about 0.2%
Radiation (mSv)	1.264	1.259	Replica lower
Temperature (°C)	39.9	39.9	Tie, shared physics path
Moonquakes / day	1.32	1.32	Tie, shared physics path
Micrometeorites	0.0	0.606	Not an independent win
Dust (g cm^{-3})	0.543	0.543	Tie, shared physics path

Table 3: Window A RMSE, 17 June to 14 July 2026, predominantly mixed, 84 site-days.

5.3.2. *Window B, predominantly lunar day*

Window B uses the same sites, blend, adaptive gating, and metrics on 18 July to 14 August 2026 (84 site-days). About 76 percent of the site-days are on the dayside. For micrometeorites, LIPAS carries a shower-timing term, including the Perseids in early August, and the replica uses the seasonal MEM 3 envelope without that term.

Solar activity is lower for the hybrid (4.93 versus 5.71, about 13.7%). Radiation is not: in this predominantly dayside window the replica is 0.0040 and the hybrid matches its physics path at 0.0259. Temperature, moonquakes, and dust remain ties (Table 4). The label is the evaluation series, not a surface impact measurement. The mission simulation later injects one scripted storm for warning-time comparison only; it is not a substitute for the disturbed Window C interval.

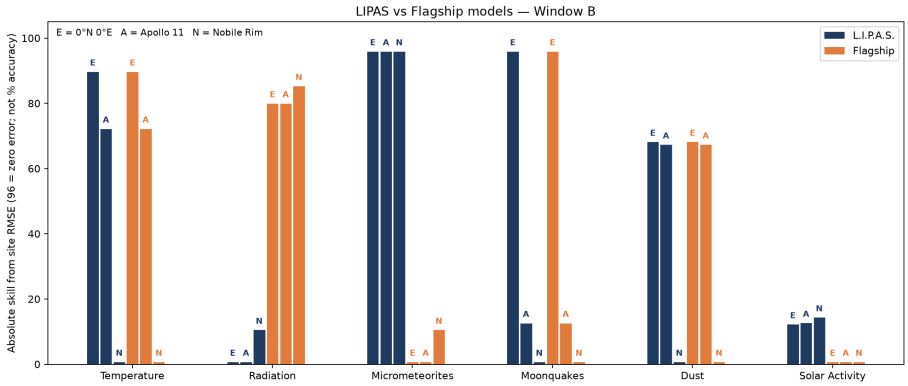


Figure 9: Window B skill from per-site RMSE (18 July to 14 August 2026; 84 site-days). Same site letters and replica as Window A.

Hazard	Hybrid	Replica	Outcome
Solar activity	4.93	5.71	Hybrid lower by about 13.7%
Radiation (mSv)	0.0259	0.0040	Replica lower (dayside window)
Temperature (°C)	24.7	24.7	Tie, shared physics path
Moonquakes / day	0.776	0.776	Tie, shared physics path
Micrometeorites	0.0	0.663	Shower term only
Dust (g cm^{-3})	0.167	0.167	Tie, shared physics path

Table 4: Window B RMSE, 18 July to 14 August 2026, lunar-day dominant, 84 site-days.

5.3.3. *Window C, disturbed historical interval*

Window C covers 4 to 17 May 2024 at the same three sites (42 site-days). Space-weather drivers are elevated relative to the quietest 2026 intervals. Storm-aware adaptive gating zeros the solar and radiation machine-learning weights when the regime is storm-like or when live radiation input is at or above 0.20 mSv. Solar activity therefore ties across hybrid, physics, and replica (RMSE 18.62). Radiation remains replica-led (0.214 versus 0.194). Temperature, moonquakes, and dust tie on the shared physics path; micrometeorite scores remain circular under blend weight zero.

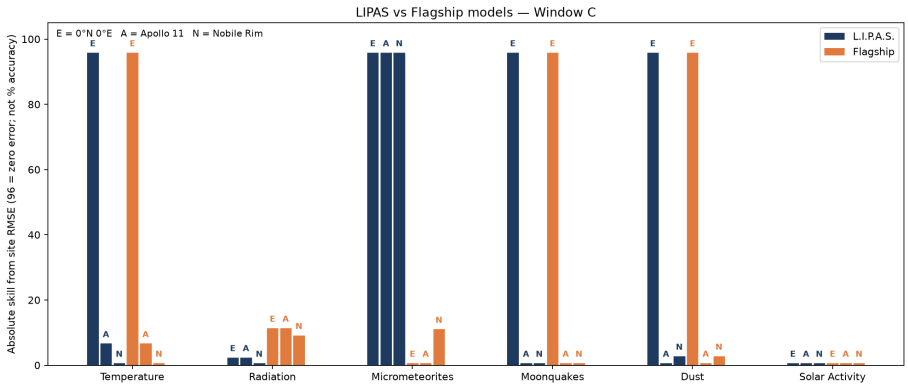


Figure 10: Window C skill from per-site RMSE (4 to 17 May 2024; 42 site-days). Same site letters and replica as Windows A and B.

Hazard	Hybrid	Replica	Outcome
Solar activity	18.62	18.62	Tie (adaptive gate to physics)
Radiation (mSv)	0.214	0.194	Replica lower (disturbed interval)
Temperature (°C)	44.34	44.34	Tie, shared physics path
Moonquakes / day	1.29	1.29	Tie, shared physics path
Micrometeorites	0.0	0.622	Shower term only
Dust (g cm^{-3})	0.549	0.549	Tie, shared physics path

Table 5: Window C RMSE, 4 to 17 May 2024, disturbed historical interval, 42 site-days.

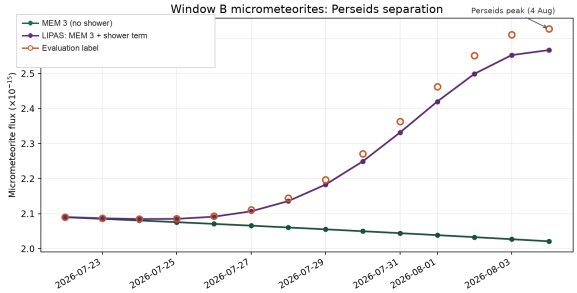


Figure 11: Window B study-site mean micrometeorite flux. LIPAS includes the shower-timing term. The comparison curve is MEM 3 without it.

5.4. Operational mission simulation

A radiation-focused EVA at the Apollo 11 site was simulated over 7 h 37 min, the duration of the second Apollo 17 EVA, to compare LIPAS warning times with those of separate public products. The same storm was injected on every path: a precursor at T+40, rising Kp and southward IMF Bz at T+55, and a GOES-like proton surge at T+70. Every path used the same thresholds, 0.095 mSv h-1 for caution and 0.15 mSv h-1 for no-go. The GOES-only path used proton flux alone, and the combined NASA path took the earlier of the proton warning and the geomagnetic or solar-driver warning.

Figure 12: dashed lines mark the precursor, geomagnetic change, and proton surge. LIPAS issued caution at T+65 and no-go at T+90. GOES alone reached caution at T+80 and no-go at T+105, and the combined path reached caution at T+55 and no-go at T+70. The earlier LIPAS caution came from upstream solar and geomagnetic drivers, not from an event the proton product could not see. The exercise is radiation-focused. It does not show that LIPAS is safer on every hazard, and it does not replace an operator or flight-certified products.

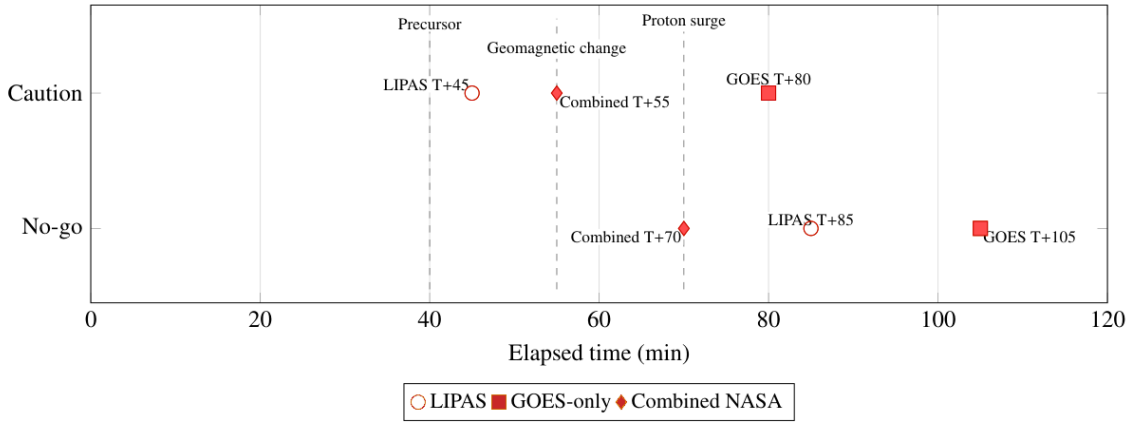


Figure 12: Mission-simulation decision timeline. Open circles are LIPAS, squares are GOES-only, and diamonds are the combined NASA path.

5.5. Input ablation

The ablation study keeps the locked model fixed and uses the quiet 5–18 July 2026 subset of Window A (42 site-days). One input at a time, and two input pairs, are replaced with their subset means; the model is not retrained. Tables 6 and 7 report the resulting RMSE values. In the full-minus-case column, a negative value means that the altered configuration had a higher error than the full model. On the radiation channel, replacing the live solar input with its subset mean raised RMSE from 0.1173 to 0.1196 (about 1.9%). Replacing any other single input left radiation RMSE essentially unchanged, and replacing solar and radiation together raised it to 0.1196 (about 2.0%). On the solar channel, the same replacement lowered RMSE from 5.142 to 3.171 (about 18.9%). This is a sensitivity result for a quiet window and does not show that live solar data are generally harmful; Section 6 discusses possible explanations. The other single and paired replacements changed solar RMSE by about 1% or less.

Configuration	RMSE	Full minus case	Relative change
Full model	0.1173	0.0000	0.0%
Set solar to mean	0.1196	−0.0023	−1.9%
Set radiation to mean	0.1151	+0.0022	+1.9%
Set temperature to mean	0.1173	∼0	∼0%
Set moonquakes to mean	0.1173	∼0	∼0%
Set micrometeorites to mean	0.1173	∼0	∼0%
Set dust to mean	0.1173	∼0	∼0%
Set solar and radiation to mean	0.1196	−0.0023	−2.0%

Table 6: Radiation-channel sensitivity results from input ablation. Changes are full-model RMSE minus configuration RMSE.

Configuration	RMSE	Full minus case	Relative change
Full model	5.142	0.000	0.0%
Set micrometeorites to mean	5.134	+0.008	+0.15%
Set radiation to mean	4.931	+0.211	+4.1%
Set moonquakes to mean	5.115	+0.027	+0.52%
Set temperature to mean	5.162	+0.020	+0.39%
Set dust to mean	5.175	−0.033	−0.64%
Set solar to mean	3.171	+0.971	+18.9% (lower RMSE)
Set moonquakes and dust to mean	5.183	−0.041	−0.81%

Table 7: Solar-activity sensitivity results from input ablation. Changes are full-model RMSE minus configuration RMSE.

6. Discussion

The studies reported in Section 5 show that the usefulness of machine learning, physics-based modelling and data integration in LIPAS depends on the hazard being forecast and on the purpose for which the output is used. This section interprets those results, sets out what they do and do not show, and identifies the limitations that bound the answers to the research questions given in Section 8. Combining physics and machine learning is not itself new (Karpatne et al., 2017; Willard et al., 2022; Camporeale, 2019). Theory-guided and physics-informed approaches have already been applied across environmental and space-weather problems. The contribution of LIPAS is narrower: the application of hazard-specific weighting, uncertainty-aware blending and cross-hazard inputs within a single lunar forecasting and decision-support system. In LIPAS the Python physics-estimation function supplies a physically grounded value for each hazard, the learned correction from the LSTM trunk is blended with it through Eqs. (2) and (3), and public heliophysics data enter as inputs to both paths rather than as a third modelling method. The results provide preliminary support for this design, while also showing that the architecture should not be treated as a replacement for specialised scientific models or as a fully validated operational forecasting system. The development record is important for interpreting the final architecture, and it must be read separately from the quantitative evidence. The April diagnostics used the browser prototype, not the locked hybrid model, and compared it with weakly replicated baselines at a single daytime site. The prototype scored higher than those weak replicas on five of the six hazards, but that comparison was designed to expose failure modes rather than to measure skill, and its scores are not pooled with the locked-model results. Its actionable finding was the temperature overshoot against the Diviner-style dayside relation near peak heating, together with run-to-run variation in physically constrained channels. These observations motivated the migration to a server-assisted backend and the decision to keep temperature, moonquakes, micrometeorites and dust on the physics path. Every quantitative claim in Windows A, B, and C, the mission simulation and the ablation therefore rests on the locked production model from the second training run. The ties on temperature, moonquakes and dust follow directly from how the system and the evaluation were constructed. The locked serve baseline blend weights are [0.4, 0.0, 0, 0, 0, 0] for solar activity, radiation, temperature, moonquakes, micrometeorites and dust, with storm-aware gating that zeros solar and radiation machine learning under storm-like regimes or when live radiation input is at or above 0.20 mSv. With a weight of zero, the hybrid output for those channels is the physics-estimation output. The replicated public baselines and the evaluation labels for the same channels were built from the same public archives and formulations. Identical RMSE values (39.9, 1.32 and 0.543 in Window A; 24.7, 0.776 and 0.167 in Window B) are therefore expected by construction. They show that the hybrid did not introduce unnecessary neural corrections where physics already provides a strong foundation, but they are not evidence of skill over the references and they are not independent validation. More generally, much of the data and many of the approaches used to build the comparison targets overlap with those used to build LIPAS itself. Where that overlap is large, agreement between LIPAS and its references partly measures internal consistency rather than predictive accuracy. This is also why the uncertainty-guided weighting matters. LIPAS does not simply average a machine-learning prediction with a physics estimate. Within each channel's locked base weight, Eq. (3) reduces the learned contribution when the Monte Carlo dropout spread widens, shifting the forecast toward the physics path when the network is less certain. The purpose of the architecture was not to force machine learning into every channel, but to allow different sources of information to carry different levels of influence. Solar activity provides the clearest evidence that the learned correction and heliophysics inputs can add predictive information. It is the channel with the largest learned weight (0.4), and the hybrid RMSE was lower than the replica in Window B (4.93 versus 5.71) and only marginally in the mixed Window A (14.15 versus 14.17); Window C ties on solar because adaptive gating routes that channel through physics. Because the gain appears in both lunar regimes rather than in only one, it is less likely to be an artefact of a single window. It remains modest, however, and it is based on three locked windows at three sites (84 site-days in A and B, 42 in the disturbed Window C). Radiation did not improve. The replica had lower RMSE in Window A (1.26 versus 1.26 for the hybrid), in the predominantly dayside Window B (0.0040 versus 0.0259), and in disturbed Window C (0.194 versus 0.214). In Window B both errors are small in absolute terms, but the replica is still clearly ahead. The served baseline weight on radiation is zero (0.0, formerly 0.05), and the present studies do not isolate whether the gap arises from the learned correction, from the radiation physics path and its inputs, or from the construction of the evaluation labels against which both paths were scored. The radiation channel should therefore be read as a current weakness of LIPAS rather than as a demonstrated strength.

The micrometeorite results need equally careful reading. The channel has a blend weight of zero, so the hybrid output is the physics path, which combines the MEM 3 export with AMS-based shower modulation. The evaluation series used for scoring carries the same shower term. The hybrid RMSE of 0.0 in all three windows is therefore circular and is not an independent win; the replica errors (0.606 in Window A, 0.663 in Window B, and 0.622 in Window C) measure the absence of the shower term rather than a deficiency of the MEM 3 framework. The separation between the two curves as the Perseids approached in Window B illustrates the effect of adding shower timing to a seasonal background model. It does not demonstrate superiority over the complete NASA micrometeorite modelling framework, and it is not validated against measured lunar impact rates. The mission simulation shows why predictive performance and operational usefulness are not the same thing. In the radiation-focused EVA scenario, LIPAS issued caution at T+65, twenty-five minutes after the injected precursor at T+40 and five minutes before the proton surge at T+70, and reached no-go at T+90. The GOES-only path reached caution at T+80 and no-go at T+105, while the combined NASA path reached caution at T+55 and the earlier hard no-go at T+70. The earlier LIPAS caution came from upstream solar and geomagnetic drivers, not from an event that the proton product could not see. The result should not be interpreted as LIPAS detecting a storm that NASA could not detect. Instead, it shows that different pathways produce different warning times depending on which signals they monitor and how those signals are mapped to thresholds. The scenario used a single injected storm, predefined caution and no-go thresholds, and one hazard. It did not measure usability with mission personnel, it does not show that LIPAS is safer on every hazard, and it is not a proof of operational safety. A unified system can also concentrate risk if its data feeds or service fail, which a set of independent products partly avoids.

The ablation study provides evidence of cross-hazard sensitivity inside the locked model, but not of causation. It used a relatively quiet portion of Window A and replaced one input at a time with its window mean, without retraining. On the radiation channel, replacing the solar input raised RMSE from 0.1173 to 0.1196 (about 1.9%), and replacing solar and radiation together raised it to 0.1196, while other single replacements left it essentially unchanged. On the solar channel, the same replacement lowered RMSE from 5.142 to 3.171, while the other replacements changed solar RMSE by about 1% or less (for example, replacing radiation changed it by 0.211, or 4.1%). Several explanations are possible for the solar result: short-term variation in the live solar input may not match the smoother evaluation labels, the current weighting may give that input too much influence under quiet conditions, or the construction of the labels may favour a smoother representation of the underlying signal. The ablation measures how far the output moves when one input is held at its average; it does not show that one hazard causes another, and the lower solar error does not show that live solar data are generally harmful. Looking across the full set of results, different hazards benefit from different forms of information. Temperature, moonquakes and dust remained on their physics paths, and their agreement with the references reflects that shared construction. Solar activity showed modest evidence that the learned correction and cross-hazard inputs can add predictive value. Radiation did not, despite its dependence on solar context in the ablation. Micrometeorite behaviour was determined by explicit shower timing rather than by machine learning. LIPAS therefore does not behave as six identical machine-learning models; it acts as a framework that assigns different levels of influence to different sources of information depending on the hazard. Several limitations affect the strength of these conclusions. First, continuous co-located lunar-surface measurements do not exist for all six hazards, so quantitative targets are evaluation labels constructed from public observations, physical formulations, historical catalogs and proxies. The results describe comparative behaviour within this evaluation framework, not accuracy against complete lunar-surface truth. Second, except for the available MEM 3 engineering-model export, the comparisons are replicas: author-built reconstructions of publicly documented methods and data, not executions of agency operational software. Their value is methodological comparability under shared inputs, dates and locations, and they should not be read as runs of the agencies' own systems. Third, the qualitative comparison in Section 5.2 is a criteria matrix, not a ranking, and it does not say that LIPAS is more accurate than any specialist archive or model. Fourth, the quantitative evidence covers three locked windows, three sites and one locked model; Window C supplies a disturbed historical interval, while the 2026 windows include mixed and predominantly dayside conditions. The strongest conclusion is therefore not that LIPAS replaces specialised NASA or NOAA models. Its value lies in providing a common forecasting and decision-support layer in which established physics, public heliophysics inputs, cross-hazard relationships and uncertainty estimates can be considered together. The results support this concept as a promising direction, while showing that substantial validation is still required before operational use.

7. Future work

LIPAS already provides a multi-hazard decision-support system for lunar operations, but further work is needed before it can be considered a trusted operational forecasting tool. The locked evaluation windows, mission simulation and ablation study have clarified where that work should focus. Rather than repeating the evaluations already completed, future work should build directly on their findings.

7.1. Radiation across activity regimes

The radiation channel is the highest priority, because the locked hybrid did not outperform the replicated public baseline in any reported window. Window C begins storm-interval RMSE reporting, but radiation still trails the replica there and in the 2026 windows. Future work should extend radiation evaluation across additional disturbed periods and identify which component of the radiation path accounts for the current gap. Potential improvements include denser OMNI, DONKI and SWPC sampling, improved treatment of protonflux events, calibrated uncertainty estimates for radiation alerts, and validation against additional historical space-weather events. Differences between LIPAS and the combined NASA warning path should be examined transparently rather than resolved by retrospective threshold tuning.

7.2. Solar input behaviour

The solar ablation result deserves closer investigation. Repeating the ablation under stronger solar activity, and with retrained rather than mean-substituted configurations, would show when the live solar input provides the greatest benefit and whether its influence under quiet conditions should be reduced. The modest solar gain should also be tested over longer periods and additional sites.

7.3. Physics-tied channels and independent labels

Temperature, moonquakes, dust and micrometeorites should remain physics-led, but their evaluation needs targets that are independent of the physics path. Future work should strengthen meteorshower timing and MEM 3 integration, and should score micrometeorites against observations that do not already contain the LIPAS shower term, so that the channel can be assessed rather than only described.

7.4. Validation with measured data

As data become available from Artemis missions, lunar landers, rovers and orbital instruments, evaluation labels should be replaced with direct measurements wherever possible. Where measurements are not yet available, estimated labels should continue to be identified clearly. Collaboration with NASA, NOAA, ESA, JAXA and university research groups would strengthen validation through access to additional datasets, operational expertise and missionrelevant performance criteria.

7.5. Operational development

Future development of the LIPAS interface and mission planner should improve uncertainty visualisation, mission-specific alert thresholds and reports aligned with existing operational checklists. Mission simulations should extend beyond a single radiation storm to multi-hazard scenarios, should test behaviour when data feeds fail, and should measure usability with mission personnel. Interface improvements are valuable only if they communicate the underlying scientific results accurately and transparently.

7.6. Extending the framework beyond the lunar surface

Because much of the system already relies on heliophysics data, extending the framework to cislunar space and Earth orbit is a logical next step, with Mars as a longerterm possibility. These environments offer substantially more observational data than the lunar surface, allowing more rigorous validation of the hybrid method. The multi-hazard architecture and channelspecific weighting could be reused without a complete redesign. The immediate priorities are therefore to diagnose and improve the radiation channel, explain the solar ablation behaviour, and obtain independent evaluation labels for the physics-tied channels. Over the medium term, measured data and collaboration with space agencies should improve confidence in the system. Throughout its development, LIPAS should remain a consultative decision-support platform designed for incomplete observations, used alongside NASA and NOAA products rather than as a replacement for specialised scientific models or flightqualified systems.

8. Conclusion

This research examined whether machine learning can be integrated into a unified, physics-based lunar forecasting system while retaining the physical foundations of established environmental models. Regarding the first research question, a hybrid lunarforecasting system can be developed by combining machine-learning and physics-based approaches. The development process showed that a hazard-specific blend was more appropriate than applying machine learning uniformly: the prototype diagnostics exposed instability in physically constrained channels, which led to a locked hybrid model in which solar activity retains a learned weight while radiation follows the physics path in the locked serve policy; temperature, moonquakes, micrometeorites and dust remain on the physics path. The second research question asked whether integrating multiple hazards and data sources into one decision-support system can support operational decision-making alongside existing public products. The radiation-focused mission simulation answered it partially: LIPAS issued caution earlier than the GOES-only path because it drew on upstream solar and geomagnetic drivers, but the combined NASA path reached the hard no-go first, and the study did not establish that LIPAS is safer than existing approaches. The third research question asked whether the locked hybrid improves prediction performance relative to replicated public baselines and physics-only references. The answer is mixed. Solar activity improved modestly in Window B and only marginally in Window A, with Window C tying on solar under adaptive gating; radiation did not; temperature, moonquakes and dust tied because they share a physics construction with their references; and the micrometeorite contrast, under blend weight zero, describes the shower term rather than independent skill. The principal contribution of LIPAS is therefore not the replacement of existing lunar models but their integration into a single forecasting and decision-support framework. The studies show both where a constrained hybrid architecture can add predictive value and where it does not yet do so, and they define the validation that must follow before LIPAS can support lunar operations alongside NASA and NOAA products.

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CRediT author statement

Luka Takki: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing: original draft, and Writing: review and editing.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

Public sources are listed in the references; numerical tables accompany the paper. Dashboard: https://lipas.netlify.app. Code and locked model weights: https://github.com/Luka-bot-afk/LIPAS (numerical tables and `locked_saved_model` in the papers/appendix_numerical_tables folder).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Cursor and Google Gemini to review, and to help modify, selected parts of the code and the manuscript. These tools did not write the paper. The author checked every suggested change, made the final edits, and takes full responsibility for the content of the published article. Their role was limited to review and selective modification of existing material; they were not used to draft or author the manuscript.

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The reported experiments are reproduced in the Supplementary Appendices. Code and study scripts: https://github.com/Luka-bot-afk/LIPAS. Daily tables and the locked model weights are in papers/appendix_numerical_tables/ (see the reproduction appendix).

Appendices

Appendix A. Numerical tables

The short RMSE tables for Windows A, B, and C are in `reported_window_rmse.md`. The daily rows are in the numerical PDF. If the preview is blank, the Download option opens the file. April rows start after the cover page. Window A daily rows start at the 28-day mixed-interval heading. Window B daily rows start at the 18 July table. Window C daily rows cover 4–17 May 2024.

- https://github.com/Luka-bot-afk/LIPAS/blob/main/papers/appendix_numerical_tables/reported_window_rmse.md
- https://github.com/Luka-bot-afk/LIPAS/blob/main/papers/appendix_numerical_tables/LIPAS_appendix_numerical_tables_A_B_D_E.pdf

Appendix B. Qualitative comparison framework

DONKI, Diviner, LDEX, Apollo PSE, MEM 3, GOES-18, and LIPAS are scored on the nine criteria. The labels run from None to Very High. Labels are not converted into numbers or summed. Scores can be compared with Table 2. The file also lists the documents used for each score.
https://github.com/Luka-bot-afk/LIPAS/blob/main/papers/appendix_numerical_tables/LIPAS_qualitative_comparison_framework.md

The nine criteria

Hazard coverage, fusion across hazards, forecast lead, spatial detail, real-time use, robustness, validation, usability, and openness.

Documents used

- NASA DONKI API documentation, https://api.nasa.gov/DONKI/
- NOAA SWPC data services, https://services.swpc.noaa.gov/
- Paige et al. (2010), Space Science Reviews, Diviner.
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Appendix C. How the numbers are compared

Sites and windows

Sites are 0°N 0°E (E), Apollo 11 (A), and Nobile Rim (N). Window A is 17 June to 14 July 2026 (84 site-days, predominantly mixed). Window B is 18 July to 14 August 2026 (84 site-days, mostly lunar day). Window C is 4 to 17 May 2024 (42 site-days, disturbed historical interval). The hazards are solar activity, radiation, temperature, moonquakes, micrometeorites, and dust.

What 96 means on the skill charts

Skill is 96 times the positive part of 1 minus RMSE divided by a hazard scale. The scale for each hazard is the larger site error, so the worst site on that hazard is 0 and a perfect site is 96. Use the same scale for LIPAS and the replica. 96 is not a percent.

The blend in this paper is solar 0.4, radiation 0.0, and 0 for temperature, moonquakes, micrometeorites, and dust, with storm-aware adaptive gating as in Section 3.2.

Appendix D. How to rerun each study

Code, configuration, data-processing helpers, and evaluation scripts for the reported studies are in the public repository: https://github.com/Luka-bot-afk/LIPAS. Step-by-step commands for each study are collected in `REPRODUCTION.md`:
https://github.com/Luka-bot-afk/LIPAS/blob/main/papers/appendix_numerical_tables/REPRODUCTION.md

The locked TensorFlow weights, scaler, and blend weights for the reported studies are shipped in the public repository at `papers/appendix_numerical_tables/locked_saved_model/`. To reproduce the Window A, Window B, mission-simulation, and ablation numbers, copy that folder to `saved_model` at the repository root (or pass `-model-dir papers/appendix_numerical_tables/locked_saved_model`) and leave the blend weights unchanged (solar 0.4, radiation 0.0, and zero for temperature, moonquakes, micrometeorites, and dust) and enable the same adaptive gating rules used in the reported runs. Expected RMSE values for each study are listed with the commands below.

1. **Qualitative comparison (Table 2).** The framework file linked in Appendix B records the scoring rules for the seven systems; labels can be compared with Table 2.
2. **April prototype (Figures 5 and 6).** The numerical PDF contains the April rows. This daytime run at 0°N, 0°E is not the locked window evaluation. There is no separate script, and these rows do not belong in Table 3 or Table 4.
3. **Window A (Table 3, Figure 7).** `python scripts/run_14day_flagship_study.py -start 2026-06-17 -end 2026-07-14 -window-label A -out-dir logs/overnight_improve/final_v3/window_A -model-dir papers/appendix_numerical_tables/locked_saved_model -csv data/ingested_extended/extended_training_window_B_ready.csv`. See `logs/overnight_improve/final_v3/window_A/summary.json`. Solar RMSE should be about 14.15 versus 14.17. Radiation should be about 1.26 versus 1.26.
4. **Window B (Table 4, Figure 9).** The same script, with start 2026-07-18, end 2026-08-14, and window label B. Solar RMSE should be about 4.93 versus 5.71. Radiation should be about 0.0259 versus 0.0040.
5. **Mission simulation (Figure 12).** `python scripts/run_mission_sim_radiation_eva.py -out-dir logs/study_mission_sim_radiation_eva`. LIPAS caution is at T+65 and no-go is at T+90.
6. **Window C (Table 5, Figure 10).** The same script, with start 2024-05-04, end 2024-05-17, and window label C. Solar RMSE should tie at about 18.62. Radiation should be about 0.214 versus 0.194.
7. **Ablation (Tables 6 and 7).** `python scripts/run_hazard_ablation_study.py -start 2026-07-05 -end 2026-07-18 -out-dir logs/overnight_improve/ablation_A_quiet`. The model is not retrained. Replacing the live solar input with its subset mean raises radiation RMSE from 0.1173 to 0.1196 and lowers solar RMSE from 5.142 to 3.171.

Appendix E. Train, validation, and test splits

Table E.8 records the chronological split used for the locked second training run. Counts come from the locked-model metadata. Dates are the common per-site range after ordering each of the twelve training sites by time. The six hazards share one multi-output split; there is no separate hazard-specific cut. Windows A, B, and C are locked evaluation sets reported in the Results and are not part of this training split.

Partition	Dates	Rows	Share
Training	1 Jan 1969 – 12 Apr 2009	176 544	≈70%
Validation	13 Apr 2009 – 28 Nov 2017	37 824	≈15%
Chronological test	29 Nov 2017 – 16 Jul 2026	37 824	≈15%
Total	1 Jan 1969 – 16 Jul 2026	252 192	100%

Table E.8: Chronological train, validation, and test split for the locked model (twelve sites; all six hazards).

Locked evaluation set	Dates	Size
Window A	17 Jun–14 Jul 2026	84 site-days
Window B	18 Jul–14 Aug 2026	84 site-days
Window C	4–17 May 2024	42 site-days

Table E.9: Locked evaluation windows used in the Results.